

A Packing of 16 Four-Spaces in $\mathbb{R}^{13}$

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Abstract

We give 16 4-dimensional subspaces of $\mathbb{R}^{13}$ whose smallest squared chordal distance is at least 2.953471. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.941951, obtained by deleting 1 of the 17 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 16 subspaces of dimension 4 in $\mathbb{R}^{13}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.953471 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 13$, $n = 4$. The cell $N = 16$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.941951, obtained by deleting 1 of the 17 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(13 - 4)/13 \cdot 16/15 = 2.953846$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m13n4N16.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 64 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
295347159808967389195106594031871225889700156499077114902347608357611599
493952987752223188688982197682087781701191734453600719894162219008700243
6196275089050121651810502761307175579840007625095
```

denominator:

```
999999999999411778455084893814773392373181120702804803771995986809676691
468084399793491620829509869300753878587763783478614109643169979820282414
745033208669777075446499778330130641981890767252
```

B Coordinates

16 blocks of 13 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m13n4N16.txt`.

```
-212004407031 -149608673738 195976966226 21743407452890510091 199041238284 3390954900 -534160223773 62322030904 83405341571 58368156192 362267981215
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